

SEET 2026 Program

Penn State Erie, The Behrend College

<u>08/09/2026</u> <u>Conference Banquet</u> <u>Venue: The Colony Family, 2670 W 8th Street, Erie, PA</u> <u>Time: 6:00 PM – 8:30 PM</u>		
<u>08/10/2026</u> <u>First Day Activities Schedule</u>		
<u>Time</u>	Conference Room: Burke REDC 180	
7:00 AM - 9:00 AM	Check-in and Continental Breakfast (In front of Clark Cafe)	
9:00 AM - 9:10 AM	Welcome Talk – Prof. Timothy Kurzweg, General Chair, SEET 2026	
9:10AM - 9:15 AM	Welcome and introduction: Prof. Ralph Ford, Chancellor of Penn State Behrend	
9:15 AM - 9:20 AM	Research at PSU – Dr. Alicyn Rhoades (Distinguished Professor of Engineering), Vice Chancellor and Associate Dean for Research and Graduate Studies	
9:20AM - 10:20 AM	Keynote: Agentic Systems: The Next Evolution of Software Engineering David Kohn AVP - Head of AI & Emerging Technologies at ValueMomentum	
10:20 AM - 10:40 AM	Group Picture (Burke REDC 180) Coffee Break (In front of Burke REDC 001/002)	
Session 1		
Time	Room: Burke REDC 001 Session Chair: Naseem Ibrahim	Room: Burke REDC 002 Session Chair: Mohammad Abid
10:40 AM – 11:00 AM	90 Unified DT–FL–DRL–Blockchain Framework for MEC Offloading Anthony Gladdney, Shashi Bhushan Jha	85 Comparative Study of Security Testing Tools for Distributed Web Applications Bouchaib Falah, Grace Julius, Onyedikachi Kanu, Chinyere Offor
11:00 AM – 11:20 AM	80 AI-Driven Autonomous Data Pipeline Orchestration in Cloud-Native Enterprise Platforms Sudhir Saxena	134 Eye-RIS: Design and Evaluation of a Low-Latency Voice-Driven AR System for Real-Time Clinical Document Analysis on Smart Glasses Robert John Luther Bennethum , Zainab Elahi Qazi, Ozge Selin Ak , Logan Swartz , Griffin Garman, Isiah Milien , Matthew Prage , Paul Koci , Sultan Al Nahian , and Sayed Mohsin Reza

11:20 AM – 11:40 AM	89 <i>Development of a Child-Friendly Constraint-Induced Movement Cast Using 3D Scanning and 3D Printing</i> Remington Orange, Bibiana M. Steckel, Christopher R. Shelton	163 <i>AI Danger Perception, Ethical Use, and Professional Regulation: A Framework for Algorithmic Alignment and Risk Management</i> Intisar Rizwan, Mohammad Atif
11:40 AM – 12:00 PM	73 <i>Data Mesh as a Migration Strategy: A Federated Approach to Cloud-Native Data Engineering</i> Richa Solanki	161 <i>An Empirical Investigation of Soft Skills and Corresponding Challenges in International Software Engineering Education</i> Sajid Ibrahim Hashmi, Jouni Markkula
Conference Room: Burke REDC 180		
12:00 PM – 01:00 PM	Keynote: Prof. Stephan Bayne Vice President for National Security & Executive Director, CISI Critical Infrastructure Security Institute Texas Tech University	
01:00 PM - 02:00 PM	Lunch (Clark Café)	
Conference Room: Burke REDC 180		
2:00 PM - 2:55 PM	Keynote: AI Shadow Dr. Mohammad Atif Farid Chief Artificial Intelligence Officer & CISO, Cyquent Inc., and Healable AI	
3:00 PM - 3:40 PM	SRC (Poster Presentation) In front of Burke REDC 180	
3:10 PM - 3:30 PM	Coffee break (In front of Burke REDC 001/002)	
Session 2		
Time	Room: Burke REDC 001 Session Chair: Sayed Mohsin Reza	Room: Burke REDC 002 Session Chair: Dusan Ramljak
3:40 PM – 4:00 PM	180 <i>Self-Healing Cloud Infrastructure: Integrating Multi-Agent Reinforcement Learning with Policy-Driven Autonomy for Resilient Enterprise Operations</i> Balasubramanian Jagannathan	140 <i>VirNetMTD: Programmable Moving Target Defense for Virtualized Networks</i> Thu Dang , Li Wang , Xing Zhou
4:00 PM – 4:20 PM	124 <i>Intrinsic Training and Compounding Accuracy: Designing AI-Assisted SAS-to-R Migration in Clinical Trials</i> David Ju and Nathalia Nascimento	114 <i>Affective and Behavioral Computing for Autism Spectrum Disorder Detection: A Survey</i> Jillienne Lapid, , Muhammad Abid, Md Selim

4:20 PM – 4:40 PM	113 <i>Machine Learning Techniques for Predominant Musical Instrument Recognition in Polyphonic Audio: A Survey of Time-Frequency Representations and Deep Learning Architectures</i> Christian Cimaglia, Muhammad Abid	162 <i>AI Turning Static Exports into Actionable Health Intelligence</i> Mohammad Atif
4:40 PM – 5:00 PM	75 <i>Resource-Efficient Deep Learning for Type 2 Diabetes Mellitus Prediction: A Comparative Case Study</i> Abdul Shaikh, Murtala Abubakar, Amira AlAmri, Naresh Adhikari and Ashhadul Islam	100 <i>CASL: A Software Framework for Gesture-Driven, Context-Aware Layout Design in Virtual Reality</i> Annika Brown, Katie Ho, Trudi Di Qi

08/11/2026	Second Day Activities Schedule	
Time	Activities (Room 180, 001, 002)	
7:30 AM – 8:30 AM	Continental Breakfast (Clark Cafe)	
SRC and PhD Track Sessions		
	Burke REDC 001 Session Chair: Mei-Huei Tang	Online Session Session Chair: Arif Ali Khan
8:30 AM – 8:50 AM	SRC-1	148: Epic-Organized vs. Requirement-Aligned Gherkin: An Empirical Evaluation of LLM-Based Acceptance Criteria Generation Shahbaz Siddeeq , Mateen Abbasi , Jussi Raskul , Zheyong Zhang, François Christophe , Tommi Mikkonen , Pekka Abrahamsson
8:50 AM – 9:10 AM	SRC-2	158: An Overview of Human Aspects in the Alignment of Requirements Engineering and Testing Afsarah Jahin, Iflaah Salman Rahul Mohanani, Burak Turhan
9:10 AM – 9:30 AM	SRC-3	168: HyQ-Guard – AI-Driven Runtime Anomaly Monitoring and Mitigation for Hybrid Quantum Workflows Ejaz Ahmed, Mikko Mäkitalo
9:30 AM – 9:50 AM	SRC-4	169: Operationalizing the EU AI Act in Software Engineering: A Systematic Mapping Study with a Focus on Healthcare

		Qaiser Khan, Abdul Raffay Saeed, Rahul Mohanan
9:50 AM – 10:30 AM	SRC-5	192: <i>Supporting Reflection on Sustainable Software Design Decisions in Software Engineering Education: An Evaluation of a Socratic Chatbot</i> Kseniia Suomalainen, Elena Cavallin, Viktoriia Olshanskaia, Maria Luce Lupetti, Jari Porras
<i>Note: Winner of the poster presentation will give an oral presentation</i>		
10:30 AM – 10:50 AM	Coffee break (In front of Burke REDC 001/002)	
Conference Room: Burke REDC 180		
10-50 AM – 11:50 AM	Keynote: Bridging the Gap Between Technical Research and the Public Sector Laura R. Dimino, Ph.D. – Assistant Director Center for Rural Pennsylvania A Legislative Agency of the Pennsylvania General Assembly	
Session 3		
Time	Burke REDC 001 Session Chair: Nathalia Nascimento	Burke REDC 002 Session Chair: Satish Mahadevan Srinivasan
11:50 AM – 12:10 PM	96 <i>Self-Adaptive Swarm-Driven Big Data Pipelines for Dynamic Workload Optimization in Spark Environments</i> Zulfiqar Ali, Israr Ur Rehman, Kashif Manzer, Muhammad Abdul Basit Ur Rahim	82 <i>Engineering and Evaluating a Machine-Learning Monitoring Pipeline for Rare-Event Early Warning: A Reference Architecture and a Four-Layer Evaluation Methodology</i> Rui Zong, Yutong Zhao, Sen He, Muhammad Abdul Basit Ur Rahim
12:10 PM – 12:30 PM	92 <i>Anchoring Large Language Models and Digital Twins in Designing Deeply Personalized mHealth Platforms</i> Muhammad Irfan Khalid, Joe Zhao, Aamir Shahzad, Muhammad Javed, Mansoor Ahmed	142 <i>A Hybrid-Simulation LoRaWAN Honeypot for Detecting RF and Protocol Attacks against IoT Networks: A Cyber-Electromagnetic Activities Perspective</i> Ali Al-Mehaizaa, Gabriele Oligeri, Faisal Al-Bishri, Adnan Akhunzada
12:30 PM – 12:50 PM	88 <i>Lightweight Data Lineage Tracing for Cross-Function Leak Detection in Serverless Applications</i> Zuhaib Nishter, Gang Li	121 <i>An Explainable Neuro-Fuzzy Multimodal Fusion Approach for Lightweight Edge-based Plant Disease Detection in Precision Agriculture</i> Muhammad Abdullah, Tanzila Kehkashan, Domagoj Tuličić, Nikola Ivković, Adnan Akhunzada

12:50 PM - 1:50 PM	Lunch - Room 100A & 100B	
Session 4		
Time	Burke REDC 001 Session Chair: Li Wang	Burke REDC 002 Session Chair: Mohammad Imran
01:50 PM – 02:10 PM	38 A Gate-Based Quantum Ising Model Approach to Solving KenKen Puzzles Using Binary Encoding Wen-Li Wang, Mei-Huei Tang, Kevin Wang, Shahid Hussain, Saif Ur Rehman Khan	27 Driving Governance and Adoption in Custom Faculty Workload System Integrations within Cloud ERP for Community Colleges Sanjiv Kumar Bhagat
02:10 PM – 02:30 PM	13 Dynamics of Artificial Intelligence and Its Impact on Modern Higher Education Rashid Khan	91 Applying Metamorphic Testing to Machine Learning: A Systematic Mapping Study A. Redempta Manzi Muneza, Kaveen Liyanage, Faqeer Ur Rehman, Emma Sheppard , Ann Marie Reinhold
02:30 PM – 02:50 PM	131 The Effectiveness of Feedback Design for Improving LLM-based Code Generation in Quantum Optimization Algorithms Venkata Sai Prathyush Turaga , Sonali Uttam Singh, Akbar Siami Namin	125 HERO: A Measurement Framework for Cost-Energy-Latency Trade-offs in Hybrid Quantum-Classical Cloud Workloads, Mr. Nirmitt Dagli, Dr Taskin Kocak Kocak Rongyu lin, Kruti Shah , Sanjeev Kumar
02:50 PM – 03:10 PM	108 Closing the Inconsistency Trap: Unified Cross-Layer Deception via Adversarial LLM Transformation Osamah Mansoor , Adeel Anjum, Madiha Haider Syed , Dr. Muhammad Talha Gul , Jie Zhao	129 Engineering Reliable Image-Forgery Localization Systems: A Multi-Dataset Study of Topology-Aware Vision Transformers Lavanya Mandava, Rohit Biswa Karma, Shanmukh Gopinath Mandava

3:10 PM – 3:30 PM	Coffee Break (In front of Burke REDC 001/002)		
Session 5			
Time	Burke REDC 001 Session Chair: Mansoor Ahmad	Burke REDC 002 Session Chair: Zulfiqar Ali	Room 207 Session Chair:
03:30 PM – 3:50 PM	98 <i>ARQM-LITE: A Small Tool for Requirements Quality Analysis</i> Dylan Porter, Satish Mahadevan Srinivasan	139 <i>VALS: A Model-Agnostic Virtual Reality Framework for Real-Time Annotation of 3D Anatomical Models</i> Pranav Balachander, Lasya Madhuri Gundapaneni, Yamini Satish1 and Khush Mistry, Evan Goldman, Sayed Mohsin Reza	141 <i>Phishing Attack Detection using Genetic Algorithms and Artificial Neural Networks</i> Ali Al-Mehaizaa , Mohammed Al-Mansoori, Mohammed Al- Mohannadi Abdullah Al- Ghushami, Adnan Akhunzada
3:50 PM – 04:10 PM	101 <i>FlowAttester: Towards a Runtime Flow Attestation Framework in Programmable NeoCloud Networks</i> Li Wang, Anbang Ruan	150 <i>Generative AI, Agentic AI, MCP, and RAG for Improving Developer Productivity</i> Rakesh Kumar, Bidanagere Nagaraju	171 <i>Triangulating Prompt Engineering Effectiveness for Edge Case Detection: Integrating Expert Quality Review with Association Rule Mining in LLM-Driven Software Testing</i> Usama Azhar, Saif Ur Rehman Khan
04:10 PM – 04:30 PM	118 <i>Indirect Feedback Alignment: A Viable Alternative for Deep Neural Networks Backpropagation</i> Jonathan Teck Meng Liaw, Pulin Agrawal, Reza Abdollahi Dezfuli Nezhad, Dheeraj Gollu	120 <i>RepoWise: A Conversational Framework for Open Source Repository Analysis with Retrieval-Augmented Generation and Structured Analytics</i> Sankalp Kashyap, Arjun Ashok, Nafiz Imtiaz Khan, Vladimir Filkov	123 <i>SA-NET: A Self-Attention Approach for Early and Accurate Skin Cancer Diagnosis Toward Intelligent Dermatology</i> Imran Ashraf, Tanzila Kehkashan, Ivan Mihaljevi, Nikola Ivkovi, Adnan Akhunzada
04:30 PM – 04:50 PM	122 <i>Scalable and Interpretable Text Intelligence for Fake News Detection</i> Imran Ashraf, Tanzila Kehkasha , Roko	157 <i>Human-Grounded AI-Assisted Software Engineering Education: An Empirical Study of Learning Outcomes</i> Junaid Shuja, Abir Saha	193 <i>An Empirical Study of How Publication Context Relates to Repository Maintenance and Community Adoption in AI Research Repositories</i>

	Milošević, Nikola Ivković, and Adnan Akhunzada		Sang Vo, Caijing Yang, Xuan Nguyen, Boshuai Ye, Arif Ali Khan
4:50 PM – 5:15 PM	Closing Ceremony		

